EXPERIMENT-2

CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB

Code:

```
clc;
clear;
close all;
%% ============================================================
% CHARACTERISTICS STUDY OF SIDEBAND POWER AND FREQUENCIES IN AM
% USING MATLAB
% ============================================================
%% 1. INITIAL PARAMETERS
Fs = 1e6;            % Sampling frequency = 1 MHz
T  = 0.01;           % Simulation time = 10 ms
t = 0:1/Fs:T-1/Fs;
% Carrier and message parameters
fc = 100e3;          % Carrier frequency = 100 kHz
fm = 10e3;           % Message frequency = 10 kHz
Ac = 1;              % Carrier amplitude
Am = 1;              % Message amplitude
R = 50;              % Load resistance = 50 ohms
%% 2. MODULATION INDEX
mu = Am/Ac;
fprintf('\n');
fprintf('=====================================================\n');
fprintf('       AM SIDEBAND POWER AND FREQUENCY STUDY\n');
fprintf('=====================================================\n');
fprintf('\nBasic Parameters:\n');
fprintf('Carrier Frequency       = %.2f kHz\n',fc/1000);
```

```
fprintf('Message Frequency       = %.2f kHz\n',fm/1000);
fprintf('Carrier Amplitude       = %.2f V\n',Ac);
fprintf('Message Amplitude       = %.2f V\n',Am);
fprintf('Load Resistance         = %.2f Ohm\n',R);
fprintf('Modulation Index        = %.2f\n',mu);
%% 3. MESSAGE SIGNAL
message = Am*cos(2*pi*fm*t);
%% 4. CARRIER SIGNAL
carrier = Ac*cos(2*pi*fc*t);
%% 5. AM SIGNAL
AM_signal = Ac*(1 + mu*cos(2*pi*fm*t)) ...  .* cos(2*pi*fc*t);
%% ============================================================
% 6. TIME DOMAIN PLOTS
% ============================================================
% Message signal
figure;
plot(t*1000,message,'LineWidth',1.2);
xlabel('Time (ms)');
ylabel('Amplitude (V)');
title('Message Signal');
grid on;
xlim([0 1]);
% Carrier signal
figure;
plot(t*1000,carrier,'LineWidth',1);
xlabel('Time (ms)');
ylabel('Amplitude (V)');
title('Carrier Signal');
grid on;
```

```
xlim([0 0.2]);
% AM signal
figure;
plot(t*1000,AM_signal,'LineWidth',1);
xlabel('Time (ms)');
ylabel('Amplitude (V)');
title('Amplitude Modulated (AM) Signal');
grid on;
xlim([0 1]);
%% ============================================================
% 7. AM SPECTRUM
% ============================================================
N = length(AM_signal);
AM_fft = fft(AM_signal);
AM_fft_shifted = fftshift(AM_fft);
f = (-N/2:N/2-1)*(Fs/N);
AM_magnitude = abs(AM_fft_shifted)/N;
figure;
plot(f/1000,AM_magnitude,'LineWidth',1.2);
xlabel('Frequency (kHz)');
ylabel('Magnitude');
title('AM Frequency Spectrum');
grid on;
xlim([70 130]);
%% ============================================================
% 8. FIND USB AND LSB FREQUENCIES
% ============================================================
f_USB = fc + fm;
f_LSB = fc - fm;
```

```
fprintf('\n');
fprintf('Sideband Frequencies:\n');
fprintf('LSB Frequency          = %.2f kHz\n',f_LSB/1000);
fprintf('Carrier Frequency      = %.2f kHz\n',fc/1000);
fprintf('USB Frequency          = %.2f kHz\n',f_USB/1000);
%% ============================================================
% 9. BANDWIDTH
% ============================================================
BW_AM = 2*fm;
fprintf('\n');
fprintf('AM Bandwidth:\n');
fprintf('BW = 2fm = %.2f kHz\n',BW_AM/1000);
%% ============================================================
% 10. CARRIER POWER
% ============================================================
Pc = Ac^2/(2*R);
fprintf('\n');
fprintf('Carrier Power:\n');
fprintf('Pc = %.6f W\n',Pc);
%% ============================================================
% 11. SIDEband POWER
% ============================================================
P_USB = (mu^2 * Pc)/4;
P_LSB = (mu^2 * Pc)/4;
P_SB_total = P_USB + P_LSB;
%% ============================================================
% 12. TOTAL AM POWER
% ============================================================
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```
P_total = Pc + P_SB_total;
%% ============================================================
% 13. POWER EFFICIENCY
% ============================================================
efficiency = (P_SB_total/P_total)*100;
%% ============================================================
% 14. DISPLAY POWER RESULTS
% ============================================================
fprintf('\n');
fprintf('Power Calculations:\n');
fprintf('Carrier Power           = %.6f W\n',Pc);
fprintf('USB Power               = %.6f W\n',P_USB);
fprintf('LSB Power               = %.6f W\n',P_LSB);
fprintf('Total Sideband Power    = %.6f W\n',P_SB_total);
fprintf('Total AM Power          = %.6f W\n',P_total);
fprintf('Transmission Efficiency = %.2f %%\n',efficiency);
%% ============================================================
% 15. MODULATION INDEX STUDY
% ============================================================
mu_values = 0.1:0.1:1.0;
P_USB_values = zeros(size(mu_values));
P_LSB_values = zeros(size(mu_values));
P_SB_values = zeros(size(mu_values));
P_total_values = zeros(size(mu_values));
Efficiency_values = zeros(size(mu_values));
for k = 1:length(mu_values)
m_index = mu_values(k);
P_USB_values(k) = ... (m_index^2 * Pc)/4;
P_LSB_values(k) = ... (m_index^2 * Pc)/4;
```

```
P_SB_values(k) = ...
P_USB_values(k) + P_LSB_values(k);
P_total_values(k) = ...
Pc + P_SB_values(k);
Efficiency_values(k) = ...
(P_SB_values(k)/P_total_values(k))*100;
end
%% ============================================================
% 16. DISPLAY MODULATION INDEX TABLE
% ============================================================
fprintf('\n');
fprintf('=====================================================\n');
fprintf('      POWER VS MODULATION INDEX\n');
fprintf('=====================================================\n');
fprintf('\n');
fprintf(' Mu     USB Power    LSB Power    Total SB     Total AM\n');
fprintf('-----------------------------------------------------\n');
for k = 1:length(mu_values)
fprintf(' %.1f     %.6f     %.6f     %.6f     %.6f\n', ...
mu_values(k), ...
P_USB_values(k), ...
P_LSB_values(k), ...
P_SB_values(k), ...
P_total_values(k));
end
fprintf('-----------------------------------------------------\n');
%% ============================================================
% 17. USB POWER VS MODULATION INDEX
% ============================================================
```

```
figure;
plot(mu_values,P_USB_values,'o-','LineWidth',1.5);
xlabel('Modulation Index (\mu)');
ylabel('USB Power (W)');
title('USB Power vs Modulation Index');
grid on;
%% ============================================================
% 18. LSB POWER VS MODULATION INDEX
% ============================================================
figure;
plot(mu_values,P_LSB_values,'s-','LineWidth',1.5);
xlabel('Modulation Index (\mu)');
ylabel('LSB Power (W)');
title('LSB Power vs Modulation Index');
grid on;
%% ============================================================
% 19. TOTAL SIDEBAND POWER VS MODULATION INDEX
% ============================================================
figure;
plot(mu_values,P_SB_values,'^-','LineWidth',1.5);
xlabel('Modulation Index (\mu)');
ylabel('Total Sideband Power (W)');
title('Total Sideband Power vs Modulation Index');
grid on;
%% ============================================================
% 20. TOTAL AM POWER VS MODULATION INDEX
% ============================================================
figure;
```

```
plot(mu_values,P_total_values,'o-','LineWidth',1.5);
xlabel('Modulation Index (\mu)');
ylabel('Total AM Power (W)');
title('Total AM Power vs Modulation Index');
grid on;
%% ============================================================
% 21. TRANSMISSION EFFICIENCY
% ============================================================
figure;
plot(mu_values,Efficiency_values,'d-','LineWidth',1.5);
xlabel('Modulation Index (\mu)');
ylabel('Efficiency (%)');
title('AM Transmission Efficiency vs Modulation Index');
grid on;
%% ============================================================
% 22. COMPARE USB AND LSB POWER
% ============================================================
figure;
plot(mu_values,P_USB_values,'o-','LineWidth',1.5);
hold on;
plot(mu_values,P_LSB_values,'s-','LineWidth',1.5);
xlabel('Modulation Index (\mu)');
ylabel('Power (W)');
title('USB and LSB Power Comparison');
legend('USB Power','LSB Power');
grid on;
%% ============================================================
% 23. AM SPECTRUM FOR DIFFERENT MODULATION INDICES
```

```
% ============================================================
mu_plot = [0.25 0.5 0.75 1.0];
figure;
for k = 1:length(mu_plot)

mu_temp = mu_plot(k);

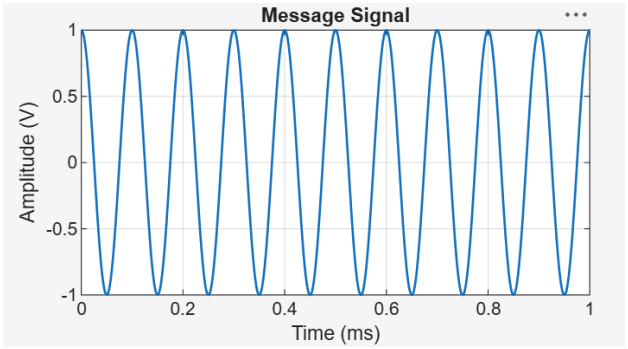
AM_temp = Ac*(1 + ...
mu_temp*cos(2*pi*fm*t)) ...
.* cos(2*pi*fc*t);
spectrum_temp = fftshift(fft(AM_temp));

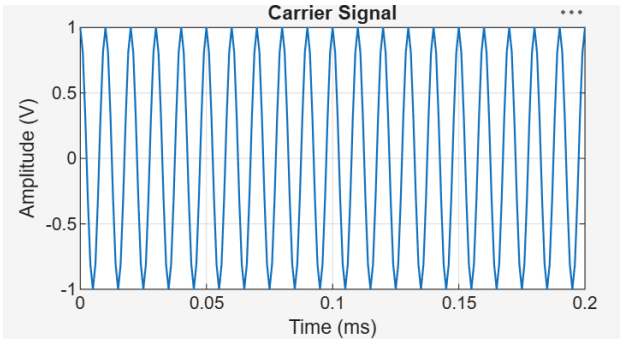
magnitude_temp = ...
abs(spectrum_temp)/N;
subplot(4,1,k);
plot(f/1000,magnitude_temp);
xlabel('Frequency (kHz)');
ylabel('Magnitude');
title(['AM Spectrum, \mu = ', ...
num2str(mu_temp)]);
grid on;
xlim([70 130]);
end
%% ============================================================
% 24. ENVELOPE OF AM SIGNAL
% ============================================================
envelope_upper = Ac*(1 + mu*cos(2*pi*fm*t));
envelope_lower = -Ac*(1 + mu*cos(2*pi*fm*t));
figure;
```

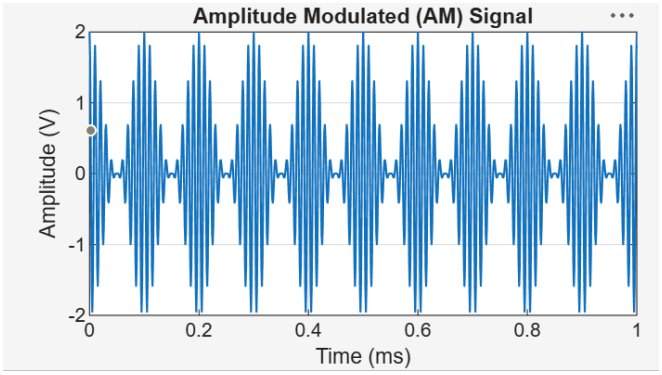
```
plot(t*1000,AM_signal);
hold on;
plot(t*1000,envelope_upper,'--','LineWidth',1.2);
plot(t*1000,envelope_lower,'--','LineWidth',1.2);
xlabel('Time (ms)');
ylabel('Amplitude (V)');
title('AM Signal and Its Envelope');
legend('AM Signal','Upper Envelope','Lower Envelope');
grid on;
xlim([0 1]);
%% ============================================================
% 25. IMPORTANT RESULTS AT 100% MODULATION
% ============================================================
mu_100 = 1;
P_USB_100 = (mu_100^2 * Pc)/4;
P_LSB_100 = (mu_100^2 * Pc)/4;
P_SB_100 = P_USB_100 + P_LSB_100;
P_total_100 = Pc + P_SB_100;
efficiency_100 = ...  (P_SB_100/P_total_100)*100;
fprintf('\n');
fprintf('=====================================================\n');
fprintf('          RESULTS FOR 100%% MODULATION\n');
fprintf('=====================================================\n');
fprintf('Carrier Power           = %.6f W\n',Pc);
fprintf('USB Power               = %.6f W\n',P_USB_100);
fprintf('LSB Power               = %.6f W\n',P_LSB_100);
fprintf('Total Sideband Power    = %.6f W\n',P_SB_100);
fprintf('Total AM Power          = %.6f W\n',P_total_100);
fprintf('Transmission Efficiency  = %.2f %%\n',efficiency_100);
```

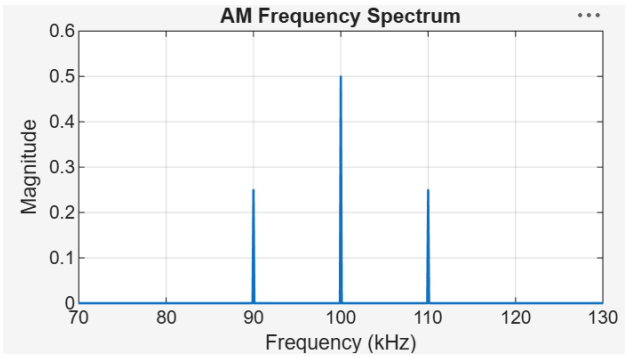
```
fprintf('=========================================================\n');
%% ============================================================
% 26. FINAL CONCLUSION
% ============================================================
fprintf('\n');
fprintf('CONCLUSION:\n');
fprintf(['1. USB frequency = fc + fm.\n']);
fprintf(['2. LSB frequency = fc - fm.\n']);
fprintf(['3. AM bandwidth = 2fm.\n']);
fprintf(['4. Sideband power increases with mu^2.\n']);
fprintf(['5. At 100%% modulation, total sideband power = Pc/2.\n']);
fprintf(['6. At 100%% modulation, total AM power = 1.5Pc.\n']);
fprintf(['7. Maximum theoretical AM efficiency = 33.33%%.\n']);
fprintf('\n=========================================================\n');
fprintf('            SIMULATION COMPLETED\n');
fprintf('=========================================================\n');
```

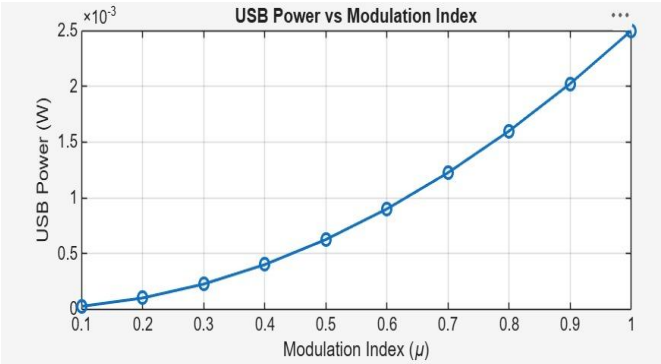
Output:

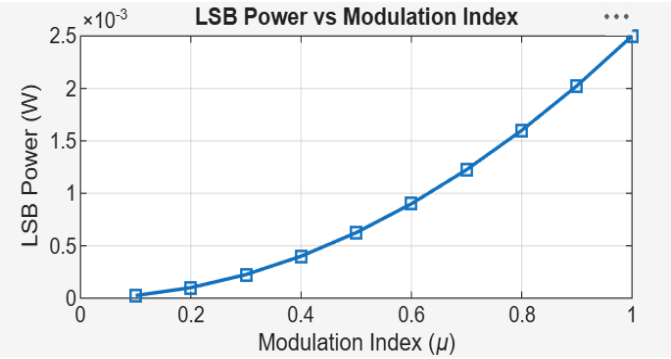


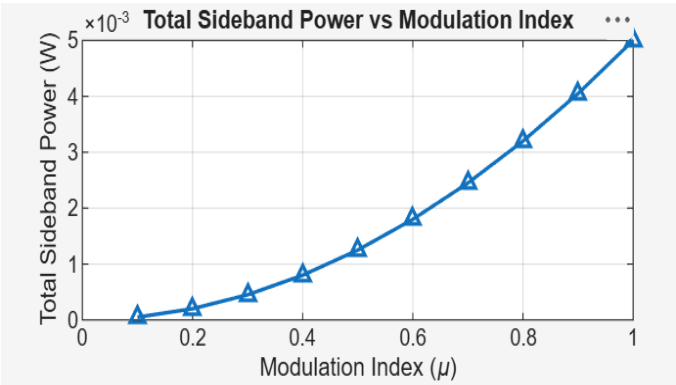


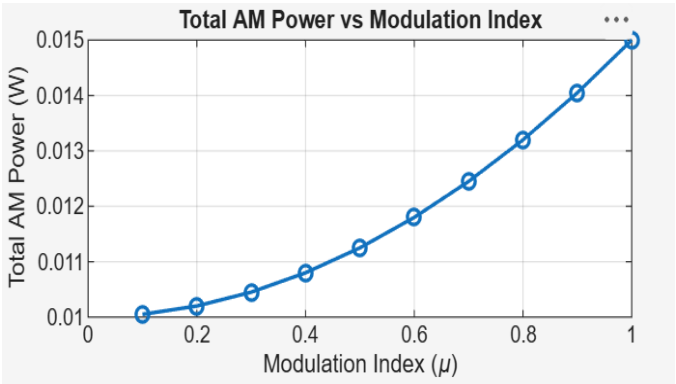


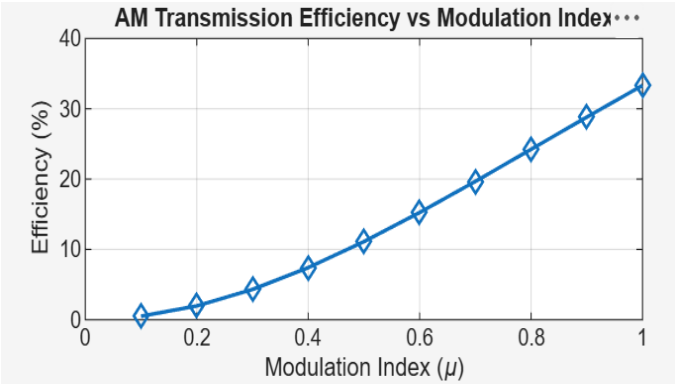


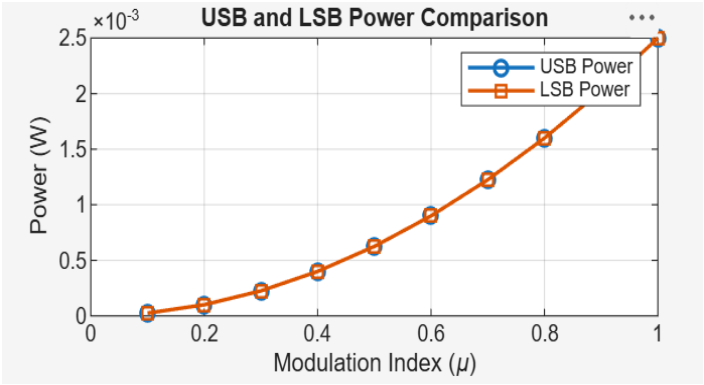

USB Power vs Modulation Index
×10-3
USB Power (W)
Modulation Index (μ)

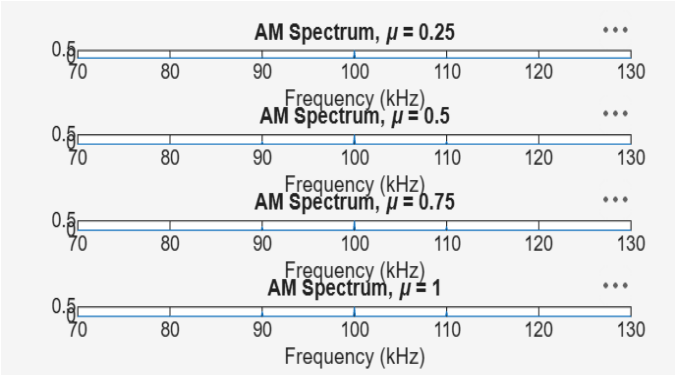

LSB Power vs Modulation Index
×10-3
LSB Power (W)
Modulation Index (μ)

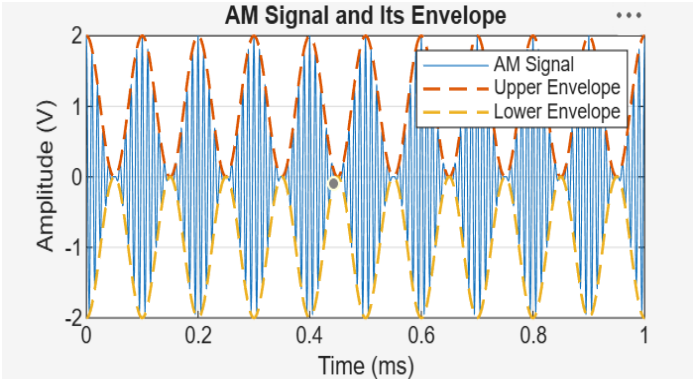

Total Sideband Power vs Modulation Index
×10-3
Total Sideband Power (W)
Modulation Index (μ)


Total AM Power vs Modulation Index
Total AM Power (W)
Modulation Index (μ)


AM Transmission Efficiency vs Modulation Index
Efficiency (%)
Modulation Index (μ)


USB and LSB Power Comparison
×10-3
USB Power
LSB Power
Power (W)
Modulation Index (μ)


AM Spectrum, μ = 0.25
AM Spectrum, μ = 0.5
AM Spectrum, μ = 0.75
AM Spectrum, μ = 1
Frequency (kHz)


AM Signal and Its Envelope
AM Signal
Upper Envelope
Lower Envelope
Amplitude (V)
Time (ms)